

Milestone 2 Evaluation Report

Group Name: ML Verifier

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Overview

This report presents the Milestone 2 evaluation of a face verification system built on the LFW dataset. The system receives two face images and produces a similarity score indicating whether they belong to the same identity, and a decision threshold converts this score into a binary prediction. Milestone 2 focuses on operating the verifier as a reproducible machine learning system. This includes tracking experiments, selecting a threshold using validation data, performing a data-centric iteration, and analyzing failure cases.

Baseline Setup and Experiment Tracking

The baseline system uses deterministic pair generation from the LFW dataset. Each pair consists of two images and a ground-truth label (0 for negative pair, 1 for positive pair) indicating whether the identities match. In our evaluation pipeline, we first load deterministic image pairs, then use these pairs to compute similarity scores (cosine or euclidean depending on config). Afterwards, we apply a decision threshold to compute verification metrics. Each run is recorded using an MLflow tracking system that logs the run identifier, dataset split, threshold rule, selected threshold, and evaluation metrics. This ensures reproducibility and allows direct comparison between runs.

Threshold Sweep and Selection Rule

A threshold sweep was performed on the validation split by keeping similarity scores fixed and varying the decision threshold. For each threshold, true positive rate (TPR) and false positive rate (FPR) were computed. Our threshold selection policy is to maximize balanced accuracy on the validation split. We used balanced accuracy (average of recall and specificity) due to the strong class imbalance between positive and negative pairs. The threshold was selected only on validation data and fixed before evaluating the test set to avoid data leakage. Using the selected threshold, the system was evaluated on the test split.

Data-Centric Iteration

In the baseline setup, exhaustive pair generation produced a validation set dominated by negative pairs. This heavy imbalance can bias threshold calibration. To address this, a deterministic sampling strategy was introduced. All positive pairs were retained, but this time, negative pairs were sampled to a 3:1 ratio, producing a more balanced validation set without affecting reproducibility. However, the test set remains unchanged across both experiments. Both systems produce identical results because they select the same optimal threshold (0.76). As a result, predictions on the test set are identical. This shows that the threshold selection process is robust to the validation sampling strategy. While the validation distribution changed, the ranking of similarity scores remained stable, leading to the same decision boundary.

Metric	Baseline	3:1 Sampled
Precision	0.0029	0.0029
Recall	0.3588	0.3588
False Positives (FP)	253,211	253,211
Balanced Accuracy	0.5345	0.5345

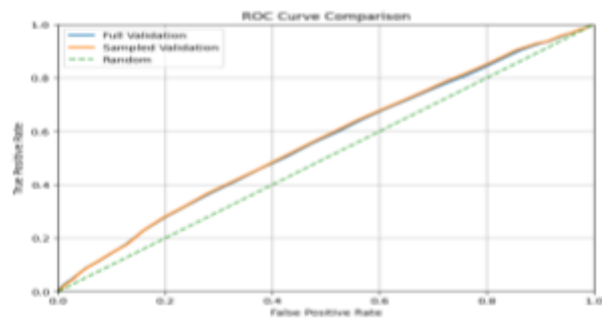


Figure 1. ROC-style plot from validation threshold sweep

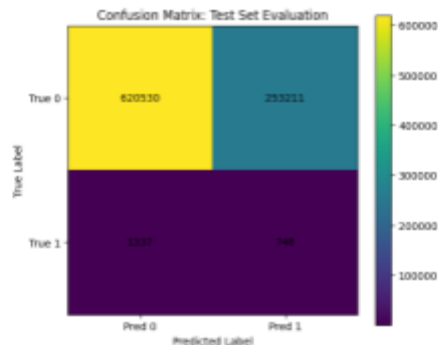


Figure 2. Confusion matrix for final test evaluation

The ROC curves for the full validation set and the sampled validation set are nearly identical. This shows that sampling preserves the underlying ranking of similarity scores while reducing the number of evaluated negative pairs. In addition, the system produces a very high number of false positives, resulting in extremely low precision (~ 0.003). While many negative pairs are correctly classified, visually similar faces often produce high similarity scores, leading to incorrect “same identity” predictions.

Error Analysis

Error Slice 1: False Negatives Near Threshold (Intra-person Variation)



Out of 1,337 false negatives, sampled examples show the same individual under different conditions (pose, lighting, expression, background). These variations reduce similarity scores below the threshold, leading to incorrect predictions. This indicates sensitivity to intra-class variation.

Error Slice 2: False Positives Near Threshold (Visually Similar Individuals)



Out of 253,211 false positives, sampled examples show individuals with similar visual features (face shape, skin tone, hairstyle, pose). These similarities produce high scores, leading to incorrect matches. Due to the large number of negative pairs, even a moderate false positive rate results in a very high FP count, causing extremely low precision.

Conclusion

This milestone demonstrates a reproducible evaluation pipeline for face verification where threshold calibration using validation data enables a principled decision boundary, and experiment tracking ensures traceability. A data-centric modification was introduced by sampling validation negatives, but analysis revealed a huge number of false positives and negatives, with the main failure arising from intra-person variation (false negatives) and visually similar individuals (false positives). Overall, the system is limited by the quality of similarity representations. We are currently comparing two ravelled ndarrays through cosine similarity and euclidean distance functions, and this is not the most effective as image data contains a lot of noise. In future milestones or pipelines, proper feature extraction techniques can be used, such as contour detection, edge detection, or removing the background to focus on the faces solely. We can also think of using a pre-trained Convolutional Neural Network such as ResNet or VGG16 to learn the features of the more accurate comparisons. However, the foundation of this milestone was successfully achieved, where we established the reproducible ML system for image verification.